Journal of Number Theory 257 (2024) 186–201

Contents lists available at ScienceDirect

Journal of Number Theory

journal homepage: www.elsevier.com/locate/jnt

General Section

The 3-rd unramified cohomology for norm one torus

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ARTICLE INFO

Article history:
Received 20 March 2023
Received in revised form 24 October 2023
Accepted 31 October 2023
Available online 24 November 2023
Communicated by L. Smajlovic

Keywords:
Unramified cohomology
Algebraic tori

ABSTRACT

For an algebraic torus S, Blinstein and Merkurjev have given an estimate of 3-rd unramified cohomology $\bar{H}^3_{nr}(F(S), \mathbb{Q}/\mathbb{Z}(2))$ obtained from a flasque resolution of S. Based on their work, for the norm one torus $W = R^{(1)}_{K/F}\mathbb{G}_m$ with K/F abelian, we compute the 3-rd unramified cohomology $\bar{H}^3_{nr}(F(W), \mathbb{Q}/\mathbb{Z}(2))$.

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1. Introduction

Let X be an integral algebraic variety over a field F, i.e. a separated scheme and of finite type over F. The i-th unramified cohomology group of X denoted by $H^i_{nr}(F(X)/F, \mathbb{Q}/\mathbb{Z}(j))$ (see section §2) defined as a subgroup of $H^i(F(X), \mathbb{Q}/\mathbb{Z}(j))$, were first introduced by Colliot-Thélène and Ojanguren [CTO89]. We will write $\bar{H}^i_{nr}(F(X)/F, \mathbb{Q}/\mathbb{Z}(j))$ for the group $H^i_{nr}(F(X)/F, \mathbb{Q}/\mathbb{Z}(j))/H^i(F, \mathbb{Q}/\mathbb{Z}(j))$.

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https://doi.org/10.1016/j.jnt.2023.10.014
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The unramified cohomology groups are homotopy invariants and can be used as an invariant to disprove the rationality of an algebraic variety. The non-triviality of the group $\bar{H}^i_{nr}(F(X)/F, \mathbb{Q}/\mathbb{Z}(j))$ implies that X is not stably rational. These groups were used to establish non stable rationality of several classes of varieties, some of them are unirational (see example [Pey93], [CTP16], [Sch19], and other results). The unramified cohomology groups also play a role on rational points. In [HSS15,HS16], a certain subgroup of the third unramified cohomology group can control the defect of weak approximation for tori over p-adic function fields.

There are no general methods to estimate the unramified cohomology groups for a general variety X. Let W be a torus over F. In 1995, Colliot-Thélène [CT95, p. 39] raised the problem: for n prime to $\mathrm{char}(F)$ and $i > 0$, determine the group $\bar{H}^i_{nr}(F(W)/F, \mu_n^{\otimes(i-1)})$. The first unramified cohomology group $\bar{H}^1_{nr}(F(W)/F, \mathbb{Z}/n\mathbb{Z})$ is trivial. The second unramified cohomology group $\bar{H}^2_{nr}(F(W)/F, \mathbb{Q}/\mathbb{Z}(1))$ coincides with the unramified Brauer group $\mathrm{Br}_{nr}(W)/\mathrm{Br}(F)$ which is isomorphic to the group $Ш^2_\omega(\hat{W})$ and has been extensively studied (see [CTS87] and [Kar87]). In 2013, Blinstein and Merkurjev investigated Colliot-Thélène's problem when $i = 3$ and obtained an exact sequence for the group $\bar{H}^3_{nr}(F(W)/F, \mathbb{Q}/\mathbb{Z}(2))$, see [BM13, Proposition 5.9].

Let K/F be a finite abelian extension and W the norm one torus $R^{(1)}_{K/F}\mathbb{G}_{m,K}$. Based on Blinstein and Merkurjev's work, we explicitly determine the p-primary component of $\bar{H}^3_{nr}(F(W)/F, \mathbb{Q}/\mathbb{Z}(2))$ which we denote by $\bar{H}^3_{nr}(F(W)/F, \mathbb{Q}/\mathbb{Z}(2))\{p\}$ for any odd prime p; furthermore, if K/F has odd degree, we completely determine the group $\bar{H}^3_{nr}(F(W)/F, \mathbb{Q}/\mathbb{Z}(2))$. To our knowledge, our result is the first one of tori whose third unramified cohomology groups are completely determined and nonzero.

Theorem. *Let $W = R^{(1)}_{K/F}(\mathbb{G}_{m,K})$ and K/F an abelian extension with $\mathrm{Gal}(K/F) = G \simeq \oplus_{i=1}^m C_i$ such that each C_i is cyclic and $\#C_{i-1}|\#C_i$ $(1 < i \leq m)$. Denote $d_i = (m-i)(m-i-1)/2$. Then for any odd prime p,*

$$\bar{H}^3_{nr}(F(W)/F, \mathbb{Q}/\mathbb{Z}(2))\{p\} \simeq H^3(G, K^*)\{p\} \oplus \oplus_{i=1}^{m-2} C_i\{p\}^{d_i}.$$

Moreover, if K/F has odd degree, then

$$\bar{H}^3_{nr}(F(W)/F, \mathbb{Q}/\mathbb{Z}(2)) \simeq H^3(G, K^*) \oplus \oplus_{i=1}^{m-2} C_i^{d_i}.$$

If $G = \mathbb{Z}/n \oplus \mathbb{Z}/n$ and n is odd, then $\bar{H}^3_{nr}(F(W)/F, \mathbb{Q}/\mathbb{Z}(2)) \simeq H^3(G, K^*)$. In particular, if F is a local field or some special number field (see Examples 3.13), then $\bar{H}^3_{nr}(F(W)/F, \mathbb{Q}/\mathbb{Z}(2)) = 0$. However, $\bar{H}^2_{nr}(F(W)/F, \mathbb{Q}/\mathbb{Z}(1)) = \mathbb{Z}/n$ is always nontrivial.

Acknowledgments. The second author is supported by National Key R&D Program of China and National Natural Science Foundation of China (Grant Nos. 12371014 and 12231009). The authors thank anonymous reviewers for their suggestions on Proposition 2.2.

2. Preliminaries

In this section, we will introduce the unramified cohomology groups. The notation and definition coincide with [BM13,CT95,CTS77].

Let F be the base field (of arbitrary characteristic) and $\Gamma = \mathrm{Gal}(F^{sep}/F)$ the absolute Galois group of F. Given an integer $j > 0$, we denote $\mathbb{Q}/\mathbb{Z}(j)$ as the direct sum of

$$\mathbb{Q}_p/\mathbb{Z}_p(j) = \begin{cases} \operatorname*{colim}_n \mu_{p^n}^{\otimes j}, & p \neq \mathrm{char}(F) \\ \operatorname*{colim}_n W_n\Omega^j_{\log}[-j], & p = \mathrm{char}(F) \end{cases}$$

in the derived category of sheaves of abelian groups on the big étale site of Spec F over all primes p, where $W_n\Omega^j_{\log}[-j]$ is the sheaf of logarithmic de Rham–Witt differentials (see [Ill79] and [BM13]).

Given a field extension L/F, let A be a rank one discrete valuation ring with fraction field L which contains F. The i-th unramified cohomology group of L/F is defined as the group

$$H^i_{nr}(L/F, \mathbb{Q}/\mathbb{Z}(j))) := \bigcap_{A \in P(L)} \mathrm{im}\ (H^i(A, \mathbb{Q}/\mathbb{Z}(j)) \longrightarrow H^i(L, \mathbb{Q}/\mathbb{Z}(j)))$$

where $P(L)$ is the set of all rank one discrete valuation rings which contains F and has quotient field L. They can also be defined by the intersection of the kernel of residue map ∂_A for $A \in P(L)$ when $\mathrm{char}(F) = 0$ [CT95].

If X is a smooth integral variety over F, the i-th unramifiel cohomology group of X is defined as $H^i_{nr}(F(X)/F, \mathbb{Q}/\mathbb{Z}(j))$. These groups are only dependent on the function field of X, and when X is proper and smooth (recall that X is smooth), we can replace $P(L)$ by the set of local rings of codimension one points in X [CT95]. If $i = 2$, the unramified cohomology group coincides with the unramified Brauer group of X. We will denote by $\bar{H}^i_{nr}(F(X)/F, \mathbb{Q}/\mathbb{Z}(j))$ the group $H^i_{nr}(F(X)/F, \mathbb{Q}/\mathbb{Z}(j))/H^i(F, \mathbb{Q}/\mathbb{Z}(j))$.

Recall that an algebraic torus of dimension n over F is an algebraic group T such that T_{sep} is isomorphic to $\mathbb{G}_m^n$. For an algebraic torus T, we write $\hat{T}_{sep}$ as the Γ-module $\mathrm{Hom}(T_{sep}, \mathbb{G}_m)$ and call it character lattice of T. The contravariant functor $T \mapsto \hat{T}_{sep}$ is an anti-equivalence between the category of algebraic tori over F and the category of Γ-lattices. We write $\hat{T}$ for the character group $\mathrm{Hom}_F(T, \mathbb{G}_m)$ and T° for the dual torus of T such that $(\hat{T}^\circ)_{sep} = (\hat{T}_{sep})^\circ$.

Let K/F be a finite Galois extension with $G = \mathrm{Gal}(K/F)$. We embed K to F_{sep}, the separable closure of F, such that the absolute Galois group Γ_K is a subgroup of Γ. We mainly consider the unramified cohomology of the norm one torus $W = R^{(1)}_{K/F}(\mathbb{G}_{m,K})$. It is defined by the equation $N_{K/F}(\Xi) = 1$ and can be constructed by the exact sequence

$$1 \longrightarrow W \longrightarrow R_{K/F}(\mathbb{G}_{m,K}) \longrightarrow \mathbb{G}_m \longrightarrow 1$$

and for its character lattice, we have

$$0 \longrightarrow \mathbb{Z} \xrightarrow{N_G} \mathbb{Z}[G] \longrightarrow \hat{W}_{sep} \longrightarrow 0$$

where N_G maps 1 to $\sum_{g\in G} g$.

A torus T is called quasisplit if its character lattice $\hat{T}_{sep}$ is a permutation Γ-module, i.e. isomorphic to the finite direct-sum of $\mathbb{Z}[\Gamma/\Gamma_i]$ for open subgroups $\Gamma_i \leq \Gamma$. And T is called flasque (resp. coflasque) if $H^1(L, \hat{T}^\circ_{sep}) = 0$ (resp. $H^1(L, \hat{T}_{sep}) = 0$) for all finite field extensions L/F. A flasque resolution (resp. coflasque resolution) of a torus T is an exact sequence of tori

$$1 \longrightarrow S \longrightarrow P \longrightarrow T \longrightarrow 1$$

(resp. $1 \longrightarrow T \longrightarrow P \longrightarrow M \longrightarrow 1$) where P is quasisplit and S is flasque (resp. M is coflasque).

Given a Γ-module A, we write for $\mathrm{Dec}(A)$ the subgroup of decomposable elements $S^2(A)$ which is generated by $(A^\Gamma)^2$ and $\mathrm{Qtr}_{\Gamma'}(A^{\Gamma'})$ for all open subgroups $\Gamma' \subset \Gamma$, with the following definition

$$\begin{aligned} \mathrm{Qtr}_{\Gamma'} :\ & A^{\Gamma'} \to S^2(A)^\Gamma \\ & a \mapsto \sum_{i<j} \sigma_i(a) \cdot \sigma_j(a) \end{aligned}$$

where $\{\sigma_i\}_i$ are the representatives for the left cosets of Γ' in Γ.

For a torus S over F, Blinstein and Merkurjev [BM13] give an estimate of $\bar{H}^3_{nr}(F(S)/F, \mathbb{Q}/\mathbb{Z}(2))$:

Theorem 2.1. *([BM13], Proposition 5.9) Let S be a torus over F and let*

$$1 \longrightarrow T \longrightarrow P \longrightarrow S \longrightarrow 1$$

be a flasque resolution of S. Then we have an exact sequence

$$\begin{aligned} 0 \longrightarrow CH^2(BT)_{tors} \longrightarrow H^1(F, T^\circ) \longrightarrow & \bar{H}^3_{nr}(F(S)/F, \mathbb{Q}/\mathbb{Z}(2)) \\ \longrightarrow & S^2(\hat{T}_{sep})^\Gamma/\mathrm{Dec}(\hat{T}_{sep}) \longrightarrow H^2(F, T^\circ). \end{aligned}$$

For an odd prime p, there is a canonical direct sum decomposition

$$\bar{H}^3_{nr}(F(S)/F, \mathbb{Q}/\mathbb{Z}(2))\{p\} \simeq H^1(F, T^\circ)\{p\} \oplus [S^2(\hat{T}_{sep})^\Gamma/\mathrm{Dec}(\hat{T}_{sep})]\{p\}.$$

The first term is the torsion subgroup of the second Chow group of the classifying variety of T.

Proposition 2.2. *Let F be a number field, then $\bar{H}^3_{nr}(F(S)/F, \mathbb{Q}/\mathbb{Z}(2))$ is a finite group for any algebraic torus S over F.*

Proof. The group $\bar{H}^3_{nr}(F(S)/F, \mathbb{Q}/\mathbb{Z}(2))$ is killed by the degree of the splitting field of S over F, and $\bar{H}^3_{nr}(F(S)/F, \mathbb{Q}/\mathbb{Z}(2))/H^1(F, T^\circ)$ is finitely generated by Theorem 2.1. Therefore, it suffices to prove that $H^1(F, T)$ is finite for any coflasque torus T, here in fact T is the dual of the flasque torus T in Theorem 2.1. Indeed, its proof is a variant of the proof of [CTS77, Theorem 1] which shows the finiteness of $H^1(F, T)$ for any flasque torus. Let K/F be a finite Galois extension which splits T, and Ω_F the set of places of F. Let $P \subset \Omega_F$ be a finite set which contains all infinite places and all ramified places in K/F and $O_P = \{a \in K | \mathrm{ord}_w(a) \geq 0 \text{ for any } v \notin P \text{ and } w|v\}$ be the ring of P-integers. We can enlarge P such that the class number of O_P is 1. This induces an exact sequence of G-modules $0 \to O_P^* \to K^* \to \mathrm{Div}\ O_P \to 0$. By construction,

$$\mathrm{Div}\ O_P \simeq \oplus_{v \notin P} \oplus_{w|v} \mathbb{Z}w \simeq \oplus_{v \notin P} \mathbb{Z}[G/G_v]$$

where $G_v = \mathrm{Gal}(K_w/F_v)$ is cyclic since v is unramified. Tensoring by $\hat{T}^\circ$ over $\mathbb{Z}$, it is still exact and hence the following is exact:

$$H^1(G, \hat{T}^\circ \otimes_{\mathbb{Z}} O_P^*) \to H^1(G, \hat{T}^\circ \otimes_{\mathbb{Z}} K^*) \to H^1(G, \hat{T}^\circ \otimes_{\mathbb{Z}} \oplus_{v \notin P} \mathbb{Z}[G/G_v]).$$

Recall that O_P is a finite generated, reduced, commutative $\mathbb{Z}$-algebra, therefore the O_P^* is of finite type (see [Roq58], [CTS77]) and the first term of above sequence is finite. The second is no other than $H^1(G, T) \simeq H^1(F, T)$. The third is 0 because G_v is cyclic and $\hat{T}^\circ$ is flasque: $H^1(G, \hat{T}^\circ \otimes_{\mathbb{Z}} \mathbb{Z}[G/G_v]) \simeq H^1(G_v, \hat{T}^\circ) \simeq H^{-1}(G_v, \hat{T}^\circ) = 0$. □

3. Main results

We continue to use the notations in §2. For the norm one torus $W = R^{(1)}_{K/F}(\mathbb{G}_{m,K})$ of Galois extension K/F, we fix a flasque resolution of W constructed from [CTS77, Proposition 15]:

$$1 \longrightarrow T \longrightarrow P \longrightarrow W \longrightarrow 1 \tag{3.1}$$

where $P = R_{K/F}(\mathbb{G}_{m,K})^r$ and r is the number of generators of $G = \mathrm{Gal}(K/F)$. We mainly consider the 3-th unramified cohomology group $\bar{H}^3_{nr}(F(W)/F, \mathbb{Q}/\mathbb{Z}(2))$ and it can be expressed by T from Theorem 2.1. Building on this work, we will reduce $\bar{H}^3_{nr}(F(W)/F, \mathbb{Q}/\mathbb{Z}(2))$ to a form determined only by the field extension K/F and the Galois group $\mathrm{Gal}(K/F)$.

Lemma 3.1. *Let $W = R^{(1)}_{K/F}(\mathbb{G}_{m,K})$ be a norm one torus of the finite Galois extension K/F with $G = \mathrm{Gal}(K/F)$ and let T be the torus in the flasque resolution (3.1) of W. Then we have $H^1(F, T^\circ) \simeq H^3(G, K^*)$.*

Proof. Firstly, we have an isomorphism $H^1(F, T^\circ) \simeq H^1(G, T^\circ(K))$ by Shapiro's lemma and Hochschild-Serre's spectral sequence. The following exact sequence is derived from (3.1):

$$1 \longrightarrow W^{\circ}(K) \longrightarrow P^{\circ}(K) \longrightarrow T^{\circ}(K) \longrightarrow 1.$$

One has $H^i(G, P^{\circ}(K)) = 0$ for $i > 0$ since $R_{K/F}(\mathbb{G}_{m,K})(K) \simeq \mathbb{Z}[G] \otimes_{\mathbb{Z}} K^*$ is an induced module. Hence we obtain an isomorphism $H^1(G, T^{\circ}(K)) \simeq H^2(G, W^{\circ}(K))$. On the other hand, the following sequence is exact

$$1 \longrightarrow \mathbb{G}_m(K) \longrightarrow R_{K/F}(\mathbb{G}_{m,K})(K) \longrightarrow W^{\circ}(K) \longrightarrow 1.$$

The proof then follows from

$$H^1(F, T^{\circ}) \simeq H^1(G, T^{\circ}(K)) \simeq H^2(G, W^{\circ}(K)) \simeq H^3(G, K^*). \quad \square$$

From the flasque resolution (3.1) of $W = R^{(1)}_{K/F}(\mathbb{G}_{m,K})$, we can obtain the following exact sequence of dual Γ-modules

$$1 \longrightarrow \hat{W}_{sep} \longrightarrow \hat{P}_{sep} \longrightarrow \hat{T}_{sep} \longrightarrow 1. \tag{3.2}$$

The part $S^2(\hat{T}_{sep})^{\Gamma}/\mathrm{Dec}(\hat{T}_{sep})$ in Theorem 2.1 is actually only dependent on the group $\mathrm{Gal}(K/F)$. To prove this, we need several lemmas.

Lemma 3.2. *([Eis13], Proposition A.2.2(d)) Let N be the kernel of the natural surjective homomorphism $S^2(\hat{P}_{sep}) \to S^2(\hat{T}_{sep})$, then the sequence*

$$0 \to \wedge^2 \hat{W}_{sep} \xrightarrow{f} \hat{W}_{sep} \otimes \hat{P}_{sep} \xrightarrow{f'} N \to 0$$

is exact where $f(a \wedge b) = a \otimes b - b \otimes a$ and f' is induced by $\hat{W}_{sep} \otimes \hat{P}_{sep} \longrightarrow S^2(\hat{P}_{sep})$.

Let g be $\hat{P}_{sep} \otimes \hat{P}_{sep} \to \hat{T}_{sep} \otimes \hat{T}_{sep}$ and g' be $S^2(\hat{P}_{sep}) \to S^2(\hat{T}_{sep})$. Because Γ_K acts trivially on $S^2(\hat{P}_{sep})$, we can only consider the G-action: $S^2(\hat{T}_{sep})^{\Gamma} = S^2(\hat{T}_{sep})^G$. The short exact sequence $0 \longrightarrow N \longrightarrow S^2(\hat{P}_{sep}) \xrightarrow{g'} S^2(\hat{T}_{sep}) \longrightarrow 0$ induces an exact sequence:

$$S^2(\hat{P}_{sep})^G \xrightarrow{g'} S^2(\hat{T}_{sep})^G \xrightarrow{\delta} H^1(G, N) \longrightarrow H^1(G, S^2(\hat{P}_{sep})). \tag{3.3}$$

Lemma 3.3. *If G is a finite group and p is an odd prime number, then*

$$H^i(G, S^2(\mathbb{Z}[G]))\{p\} = H^i(G, \wedge^2 \mathbb{Z}[G])\{p\} = 0$$

for $i \geq 1$. Furthermore, if G has odd order, then

$$H^i(G, S^2(\mathbb{Z}[G])) = H^i(G, \wedge^2 \mathbb{Z}[G]) = 0$$

for $i \geq 1$.

Proof. We have a commutative diagram with the row exact:

$$\begin{array}{ccccccccc} & & & & S^2(\mathbb{Z}[G]) & = & S^2(\mathbb{Z}[G]) & & \\ & & & & \downarrow{\scriptstyle i_1} & & \downarrow{\scriptstyle \times 2} & & \\ 0 & \longrightarrow & \wedge^2\mathbb{Z}[G] & \xrightarrow{i_2} & \mathbb{Z}[G]^{\otimes 2} & \xrightarrow{p_1} & S^2(\mathbb{Z}[G]) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \times 2} & & \downarrow{\scriptstyle p_2} & & & & \\ & & \wedge^2\mathbb{Z}[G] & = & \wedge^2\mathbb{Z}[G] & & & & \end{array}$$

where $i_1(a\cdot b)=a\otimes b+b\otimes a$, $i_2(a\wedge b)=a\otimes b-b\otimes a$, $p_1(a\otimes b)=a\cdot b$ and $p_2(a\otimes b)=a\wedge b$.

Therefore, this induces a commutative diagram with the row exact for $i\geq 1$:

$$\begin{array}{ccccc} & & H^i(G,S^2(\mathbb{Z}[G])) & = & H^i(G,S^2(\mathbb{Z}[G])) \\ & & \downarrow{\scriptstyle i_1^*} & & \downarrow{\scriptstyle \times 2} \\ H^i(G,\wedge^2\mathbb{Z}[G]) & \xrightarrow{i_2^*} & H^i(G,\mathbb{Z}[G]^{\otimes 2}) & \xrightarrow{p_1^*} & H^i(G,S^2(\mathbb{Z}[G])) \\ \downarrow{\scriptstyle \times 2} & & \downarrow{\scriptstyle p_2^*} & & \\ H^i(G,\wedge^2\mathbb{Z}[G]) & = & H^i(G,\wedge^2\mathbb{Z}[G]) & & \end{array}$$

Since $H^i(G,\mathbb{Z}[G]^{\otimes 2})=0$, one obtains $2\cdot H^i(G,S^2(\mathbb{Z}[G]))=2\cdot H^i(G,\wedge^2\mathbb{Z}[G])=0$. This implies $H^i(G,S^2(\mathbb{Z}[G]))\{p\}=H^i(G,\wedge^2\mathbb{Z}[G])\{p\}=0$ for $i\geq 1$ and odd prime p. If G has odd order, then $H^i(G,S^2(\mathbb{Z}[G]))\{p\}$ and $H^i(G,\wedge^2\mathbb{Z}[G])\{p\}$ have odd order, the proof then follows. □

Lemma 3.4. *Let P be the quasisplit torus, then $S^2(\hat{P}_{sep})^G/\mathrm{Dec}(\hat{P}_{sep})=0$.*

Proof. By the definition of Dec, we can deduce (see [BM13, A-II]):

$$S^2(A\otimes B)^\Gamma \simeq S^2(A)^\Gamma \oplus (A\otimes B)^\Gamma \oplus S^2(B)^\Gamma,$$
$$\mathrm{Dec}(A\oplus B)\simeq \mathrm{Dec}(A)\oplus \mathrm{Dec}(A,B)\oplus \mathrm{Dec}(B),$$

where $\mathrm{Dec}(A,B)$ is the subgroup of $(A\otimes B)^\Gamma$ generated by elements $\mathrm{Tr}(a\otimes b)=\sum_i \sigma_i a\otimes \sigma_i b$ for all open subgroups Γ' of Γ and all $a\in A^{\Gamma'}$ and $b\in B^{\Gamma'}$, where $\{\sigma_i\}$ is a representative of Γ/Γ'. Assume that A and B are permutation modules, it is clear that $(A\otimes B)^\Gamma=\mathrm{Dec}(A,B)$. So we can reduce to the case $\hat{P}_{sep}=\mathbb{Z}[\Gamma/\Gamma_L]$ by induction, where L/F is a finite field extension and Γ_L is the absolute Galois group of L.

We denote $\bar{\sigma}_1\cdot\bar{\sigma}_2$ to be the image of $\bar{\sigma}_1\otimes\bar{\sigma}_2$ in $S^2(\mathbb{Z}[\Gamma/\Gamma_L])$, where $\bar{\sigma}_i\in\Gamma/\Gamma_L$ and $i=1,2$. Let $<\bar{\sigma}_1\cdot\bar{\sigma}_2>$ be the orbit of Γ-action on $\bar{\sigma}_1\cdot\bar{\sigma}_2$. Let $N_{\bar{\sigma}}:=\sum_{a\in<\bar{e}\cdot\bar{\sigma}>}a$. It is clear that $S^2(\mathbb{Z}[\Gamma/\Gamma_L])^\Gamma$ is generated by all $N_{\bar{\sigma}}$, $\bar{\sigma}\in\Gamma/\Gamma_L$. It suffices to show $N_{\bar{\sigma}}\in\mathrm{Dec}(\mathbb{Z}[\Gamma/\Gamma_L])$.

Let $\Gamma'=\Gamma_L\cap\Gamma_{\sigma(L)}$ and $\Gamma_{\bar{\sigma}}$ the stabilizer of $\bar{e}\cdot\bar{\sigma}$, which is an open subgroup of Γ containing Γ'.

Lemma 3.5. *Either* $\Gamma_{\bar{\sigma}} = \Gamma'$ *or* $\Gamma_{\bar{\sigma}} = \Gamma' \cup \sigma'\Gamma'$, *where* $\sigma'\Gamma_L = \sigma\Gamma_L$ *and* $\sigma' \notin \Gamma'$.

Proof. We only need to prove the case $\Gamma_{\bar{\sigma}} \neq \Gamma'$. Assume that $\sigma', \sigma'' \in \Gamma_{\bar{\sigma}} \setminus \Gamma'$. So $\sigma'\bar{\sigma} = \sigma''\bar{\sigma} = \bar{e}$ and $\bar{\sigma}' = \bar{\sigma}'' = \bar{\sigma}$. Then ${\sigma'}^{-1}\sigma'' \in \Gamma_{\sigma(L)} \cap \Gamma_L = \Gamma'$, hence $\sigma'\Gamma' = \sigma''\Gamma'$. □

Let $\bar{\sigma} \in \Gamma/\Gamma_L$ and $\sigma_1, \cdots, \sigma_m$ representatives of Γ/Γ'. We have

$$\begin{aligned}
\mathrm{Qtr}_{\Gamma'}(\bar{e}+\bar{\sigma}) &= \sum_{i<j} \sigma_i(\bar{e}+\bar{\sigma}) \cdot \sigma_i(\bar{e}+\bar{\sigma}) \\
&= \sum_{i<j} \bar{\sigma}_i \cdot \bar{\sigma}_j + \sum_{i<j} \sigma_i\bar{\sigma} \cdot \sigma_j\bar{\sigma} + \sum_{i<j} \bar{\sigma}_i \cdot \sigma_j\bar{\sigma} + \sum_{i>j} \bar{\sigma}_i \cdot \sigma_j\bar{\sigma} \\
&= \mathrm{Qtr}_{\Gamma'}(\bar{e}) + \mathrm{Qtr}_{\Gamma'}(\bar{\sigma}) + \mathrm{Tr}(\bar{e}) \cdot \mathrm{Tr}(\bar{e}) - \sum_i \bar{\sigma}_i \cdot \sigma_i\bar{\sigma}.
\end{aligned}$$

Therefore, $\sum_i \bar{\sigma}_i \cdot \sigma_i\bar{\sigma} \in \mathrm{Dec}(\mathbb{Z}[\Gamma/\Gamma_L])$. If $\Gamma_{\bar{\sigma}} = \Gamma'$, then $N_{\bar{\sigma}} = \sum_i \bar{\sigma}_i \cdot \sigma_i\bar{\sigma} \in \mathrm{Dec}(\mathbb{Z}[\Gamma/\Gamma_L])$.

Suppose $[\Gamma_{\bar{\sigma}} : \Gamma'] = 2$. Let $\sigma' \in \Gamma_{\bar{\sigma}} \setminus \Gamma'$ and $\sigma_1, \cdots, \sigma_m$ representatives of $\Gamma/\Gamma_{\bar{\sigma}}$, hence $\sigma_1, \cdots, \sigma_m, \sigma_1\sigma', \cdots, \sigma_m\sigma'$ are representatives of $\Gamma/\Gamma_{\bar{\sigma}}$. Since $\bar{\sigma}' = \bar{\sigma} \in \Gamma/\Gamma_L$ by Lemma 3.5, one has $\bar{e} + \bar{\sigma}' = \bar{e} + \bar{\sigma} \in \mathbb{Z}[\Gamma/\Gamma_L]^{\Gamma_{\bar{\sigma}}}$, hence

$$\begin{aligned}
\mathrm{Qtr}_{\Gamma_{\bar{\sigma}}}(\bar{e}+\bar{\sigma}') &= \sum_{i<j} \sigma_i(\bar{e}+\bar{\sigma}') \cdot \sigma_j(\bar{e}+\bar{\sigma}') \\
&= \sum_{i<j} \bar{\sigma}_i \cdot \bar{\sigma}_j + \sum_{i<j} \sigma_i\bar{\sigma}' \cdot \sigma_j\bar{\sigma}' + \sum_{i<j} \bar{\sigma}_i \cdot \sigma_j\bar{\sigma}' + \sum_{i>j} \bar{\sigma}_i \cdot \sigma_j\bar{\sigma}' \\
&= \mathrm{Qtr}_{\Gamma'}(\bar{e}) - \sum_i \bar{\sigma}_i \cdot \sigma_i\bar{\sigma},
\end{aligned}$$

which implies $N_{\bar{\sigma}} = \sum_i \bar{\sigma}_i \cdot \sigma_i\bar{\sigma} \in \mathrm{Dec}(\mathbb{Z}[\Gamma/\Gamma_L])$, the proof of Lemma 3.4 then follows. □

Remark. Let $S = \mathrm{Spec}\ F$ be the trivial torus. Then $1 \longrightarrow P \longrightarrow P \longrightarrow S \longrightarrow 1$ is a flasque resolution of S. Let p be an odd prime. Therefore

$$0 = \bar{H}^3_{nr}(F(S)/F, \mathbb{Q}/\mathbb{Z}(2))\{p\} \simeq H^1(F, P^\circ)\{p\} \oplus [S^2(\hat{P}_{sep})^\Gamma/\mathrm{Dec}(\hat{P}_{sep})]\{p\}$$

by Theorem 2.1. Hence $[S^2(\hat{P}_{sep})^\Gamma/\mathrm{Dec}(\hat{P}_{sep})]\{p\}$. However, the 2-primary part cannot be determined by this method, since $H^2(F, P^\circ)$ is not in control.

Corollary 3.6. *If* p *is an odd prime number, the following homomorphism induced by the sequence* (3.3) *is an isomorphism*

$$[S^2(\hat{T}_{sep})^G/g'(\mathrm{Dec}(\hat{P}_{sep}))]\{p\} \xrightarrow{\delta} H^1(G, N)\{p\}$$

Proof. By Lemma 3.4 and Lemma 3.3, the sequence (3.3) gives the isomorphism

$$[S^2(\hat{T}_{sep})^G/g'(\mathrm{Dec}(\hat{P}_{sep}))]\{p\} \xrightarrow{\delta} H^1(G, N)\{p\}. \quad \square$$

Lemma 3.7. *If $\#[G,G]$ is a power of 2, then for an odd prime number p*

$$[S^2(\hat{T}_{sep})^{\Gamma}/\mathrm{Dec}(\hat{T}_{sep})]\{p\} \simeq [H^1(G,N)/\delta((\hat{T}_{sep}^G)^{\otimes 2})]\{p\},$$

where N, δ are defined in (3.3).

Proof. Let H be a subgroup of G, $G^{ab} = G/[G,G]$ and $H^{ab} = H/[H,H]$. Since $\#[G,G]$ is a power of 2, the exact sequence $0 \longrightarrow \mathbb{Z} \xrightarrow{N} \mathbb{Z}[G] \longrightarrow \hat{W}_{sep} \longrightarrow 0$ deduces the following commutative diagram

$$\begin{array}{ccccccc}
H^1(G,\hat{W}_{sep}) & \xrightarrow{\simeq} & H^2(G,\mathbb{Z}) & \xrightarrow{\simeq} & H^1(G,\mathbb{Q}/\mathbb{Z}) & \xrightarrow{\simeq} & H^1(G^{ab},\mathbb{Q}/\mathbb{Z}) \\
\downarrow{\scriptstyle j} & & \downarrow & & \downarrow & & \downarrow{\scriptstyle j'} \\
H^1(H,\hat{W}_{sep}) & \xrightarrow{\simeq} & H^2(H,\mathbb{Z}) & \xrightarrow{\simeq} & H^1(H,\mathbb{Q}/\mathbb{Z}) & \xrightarrow{\simeq} & H^1(H^{ab},\mathbb{Q}/\mathbb{Z})
\end{array}$$

Therefore, the cokernel $\mathrm{Coker}(j) \simeq \mathrm{Coker}(j')$ is a 2-group.

The sequence (3.2) derives the following commutative diagram with the row exact

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \hat{W}_{sep}^G & \longrightarrow & \hat{P}_{sep}^G & \longrightarrow & \hat{T}_{sep}^G & \longrightarrow & H^1(G,\hat{W}_{sep}) \longrightarrow H^1(G,\hat{P}_{sep}) = 0 \\
 & & \downarrow & & \downarrow{\scriptstyle r} & & \downarrow{\scriptstyle r'} & & \downarrow{\scriptstyle j} \\
0 & \longrightarrow & \hat{W}_{sep}^H & \longrightarrow & \hat{P}_{sep}^H & \longrightarrow & \hat{T}_{sep}^H & \longrightarrow & H^1(H,\hat{W}_{sep}) \longrightarrow H^1(H,\hat{P}_{sep}) = 0.
\end{array}$$

Since the $\mathrm{Coker}(j)$ is a 2-group and the sequence $\mathrm{Coker}(r) \to \mathrm{Coker}(r') \to \mathrm{Coker}(j) \to 0$ is exact, it shows that $2^m \hat{T}_{sep}^H \subset \mathrm{im}(\hat{P}_{sep}^H) + \hat{T}_{sep}^G \subset \hat{T}_{sep}^H$ for some integer m.

Let $\{\gamma_i\}$ be the set of representatives for the left cosets of H in G. If $a \in \hat{T}_{sep}^H$, then $2^m a = b + c \in \hat{T}_{sep}^H, b \in \mathrm{im}(\hat{P}_{sep}^H), c \in \hat{T}_{sep}^G$. By the commutative diagram

$$\begin{array}{ccc}
\hat{P}_{sep}^H & \longrightarrow & \hat{T}_{sep}^H \\
\downarrow{\scriptstyle \mathrm{Qtr}} & & \downarrow{\scriptstyle \mathrm{Qtr}} \\
S^2(\hat{P}_{sep})^G & \xrightarrow{g'} & S^2(\hat{T}_{sep})^G,
\end{array}$$

one obtains

$$\begin{aligned}
\mathrm{Qtr}(2^m a) &= \sum_{i<j} \gamma_i(b+c) \cdot \gamma_j(b+c) \\
&= \sum_{i<j} \gamma_i b \cdot \gamma_j b + \sum_{i<j} c \cdot c + lc \cdot \sum_i \gamma_i b,
\end{aligned}$$

where $l = [G:H] - 1$. So

$$\mathrm{Qtr}(2^m \hat{T}_{sep}^H) = 2^{2m} \mathrm{Qtr}(\hat{T}_{sep}^H) \subset g'(\mathrm{Dec}(\hat{P}_{sep})) + (\hat{T}_{sep}^G)^2.$$

Let H run through all subgroups of G, since G is finite, there exists a large enough integer m' such that $2^{m'}\mathrm{Dec}(\hat{T}_{sep}) \subset g'(\mathrm{Dec}(\hat{P}_{sep})) + (\hat{T}_{sep}^G)^2$, hence

$$\begin{aligned}[S^2(\hat{T}_{sep})^G/2^{2m'}\mathrm{Dec}(\hat{T}_{sep})]\{p\} &= [S^2(\hat{T}_{sep})^G/(g'(\mathrm{Dec}(\hat{P}_{sep})) + (\hat{T}_{sep}^G)^2)]\{p\} \\ &= [S^2(\hat{T}_{sep})^G/\mathrm{Dec}(\hat{T}_{sep})]\{p\}.\end{aligned}$$

On the other hand, by Corollary 3.6, the sequence (3.3) gives the isomorphism Therefore,

$$\begin{aligned}[S^2(\hat{T}_{sep})^\Gamma/\mathrm{Dec}(\hat{T}_{sep})]\{p\} &= [S^2(\hat{T}_{sep})^G/2^{2m'}\mathrm{Dec}(\hat{T}_{sep})]\{p\} \\ &\simeq [H^1(G,N)/\delta((\hat{T}_{sep}^G)^2)]\{p\}. \quad \square\end{aligned}$$

Remark. Assume that G is abelian. Then $\mathrm{Dec}(\hat{T}_{sep}) = g'(\mathrm{Dec}(\hat{P}_{sep})) + (\hat{T}_{sep}^G)^2$ from the proof above. By Lemma 3.4 and the sequence (3.3), we deduce the following isomorphism

$$S^2(\hat{T}_{sep})^G/\mathrm{Dec}(\hat{T}_{sep}) \xrightarrow{\simeq} H^1(G,N)/\delta((\hat{T}_{sep}^G)^{\otimes 2}). \tag{3.4}$$

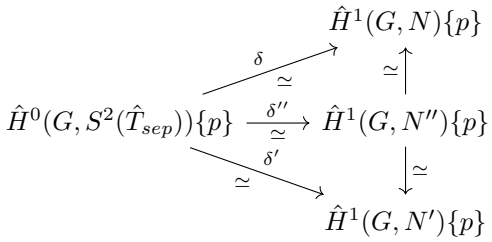
Let N' be the kernel of $\hat{P}_{sep} \otimes \hat{T}_{sep} \to S^2(\hat{T}_{sep})$ and N'' the kernel of $\hat{P}_{sep} \otimes \hat{P}_{sep} \to S^2(\hat{T}_{sep})$. Then we have the following commutative diagram with the rows exact:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & N & \longrightarrow & S^2(\hat{P}_{sep}) & \longrightarrow & S^2(\hat{T}_{sep}) & \longrightarrow & 0 \\
 & & \uparrow & & \uparrow & & \| & & \\
0 & \longrightarrow & N'' & \longrightarrow & \hat{P}_{sep} \otimes \hat{P}_{sep} & \longrightarrow & S^2(\hat{T}_{sep}) & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \| & & \\
0 & \longrightarrow & N' & \longrightarrow & \hat{P}_{sep} \otimes \hat{T}_{sep} & \longrightarrow & S^2(\hat{T}_{sep}) & \longrightarrow & 0
\end{array}$$

Taking the Galois cohomology, we can get the following commutative diagram with the rows exact:

$$\begin{array}{ccccccc}
\hat{H}^0(G, S^2(\hat{P}_{sep})) & \longrightarrow & \hat{H}^0(G, S^2(\hat{T}_{sep})) & \xrightarrow{\delta} & \hat{H}^1(G,N) & \longrightarrow & \hat{H}^1(G, S^2(\hat{P}_{sep})) \\
\uparrow & & \| & & \uparrow & & \uparrow \\
\hat{H}^0(G, \hat{P}_{sep} \otimes \hat{P}_{sep}) & \longrightarrow & \hat{H}^0(G, S^2(\hat{T}_{sep})) & \xrightarrow{\delta''} & \hat{H}^1(G,N'') & \longrightarrow & \hat{H}^1(G, \hat{P}_{sep} \otimes \hat{P}_{sep}) \\
\downarrow & & \| & & \downarrow & & \downarrow \\
\hat{H}^0(G, \hat{P}_{sep} \otimes \hat{T}_{sep}) & \longrightarrow & \hat{H}^0(G, S^2(\hat{T}_{sep})) & \xrightarrow{\delta'} & \hat{H}^1(G,N') & \longrightarrow & \hat{H}^1(G, \hat{P}_{sep} \otimes \hat{T}_{sep}).
\end{array}$$

By construction $\hat{P}_{sep} \simeq \mathbb{Z}[G]^{\oplus r}$, for $i \geq 1$ and an odd prime p, $\hat{H}^i(G, \hat{P}_{sep} \otimes \hat{P}_{sep}) = \hat{H}^i(G, \hat{P}_{sep} \otimes \hat{T}_{sep}) = 0$ by Shapiro's Lemma, $\hat{H}^i(G, S^2(\hat{P}_{sep}))\{p\} = 0$ by Lemma 3.3. Therefore the following diagram is commutative:



and we deduce

$$\begin{aligned}[H^1(G,N)/\delta((\hat{T}_{sep}^G)^{\otimes 2})]\{p\} &\simeq [H^1(G,N')/\delta'((\hat{T}_{sep}^G)^{\otimes 2})]\{p\}\\ &\simeq [H^1(G,N'')/\delta''((\hat{T}_{sep}^G)^{\otimes 2})]\{p\}.\end{aligned}$$

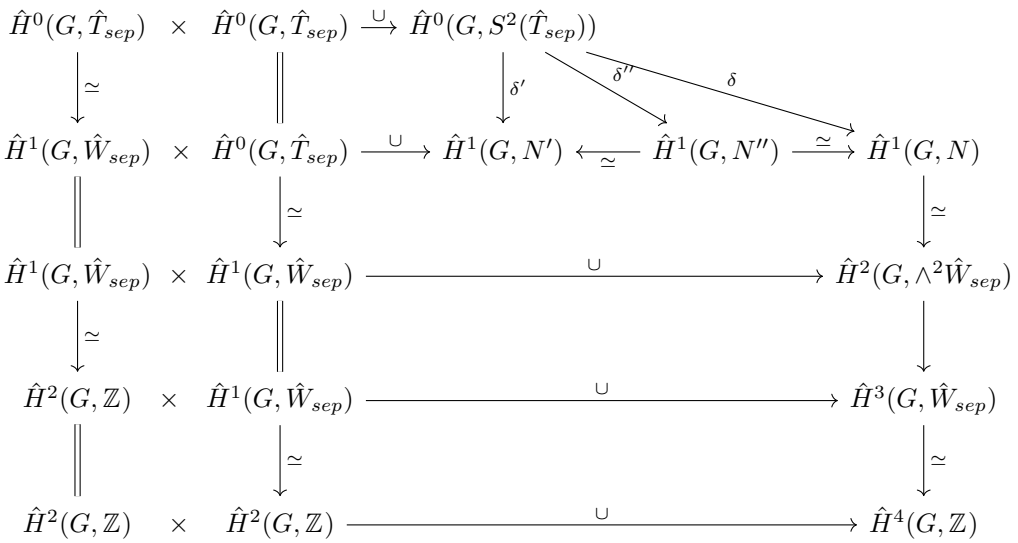
Proposition 3.8. *Let $W = R^{(1)}_{K/F}(\mathbb{G}_{m,K})$ and K/F the finite Galois extension with $G = \mathrm{Gal}(K/F)$. Let*

$$1 \longrightarrow T \longrightarrow P \longrightarrow W \longrightarrow 1$$

be the flasque resolution (3.1) of W. Suppose the commutator $[G,G]$ is a 2-group. Then

$$[S^2(\hat{T}_{sep})^\Gamma/\mathrm{Dec}(\hat{T}_{sep})]\{p\} \simeq \mathrm{Coker}(H^2(G,\mathbb{Z}) \times H^2(G,\mathbb{Z}) \xrightarrow{\cup} H^4(G,\mathbb{Z}))\{p\}.$$

Proof. By [NSW13, Chapter I, §4], we have the following commutative diagram:



a. The first square is induced by the exact sequences: $1 \longrightarrow \hat{W}_{sep} \longrightarrow \hat{P}_{sep} \longrightarrow \hat{T}_{sep} \longrightarrow 1$ and $0 \longrightarrow N' \longrightarrow \hat{P}_{sep} \otimes \hat{T}_{sep} \longrightarrow S^2(\hat{T}_{sep}) \longrightarrow 0$. Note that $\hat{H}^0(G,\hat{P}_{sep}) = \hat{H}^1(G,\hat{P}_{sep}) = 0$ thus the vertical arrows at left is an isomorphism.

b. The second square is induced by the exact sequences: $1 \longrightarrow \hat{W}_{sep} \longrightarrow \hat{P}_{sep} \longrightarrow \hat{T}_{sep} \longrightarrow 1$ and $0 \longrightarrow \wedge^2\hat{W}_{sep} \xrightarrow{f} \hat{W}_{sep} \otimes \hat{P}_{sep} \longrightarrow N \longrightarrow 0$ from the assumption and Lemma 3.2. Note that $\hat{H}^1(G, \hat{W}_{sep} \otimes \hat{P}_{sep}) = \hat{H}^2(G, \hat{W}_{sep} \otimes \hat{P}_{sep}) = 0$.

c. The third square is induced by the exact sequences: $0 \longrightarrow \mathbb{Z} \xrightarrow{N_G} \mathbb{Z}[G] \longrightarrow \hat{W}_{sep} \longrightarrow 0$ and $0 \longrightarrow \hat{W}_{sep} \xrightarrow{\varphi} \wedge^2\mathbb{Z}[G] \longrightarrow \wedge^2\hat{W}_{sep} \longrightarrow 0$ from the definition and the following Lemma 3.9.

d. The fourth square is induced by the exact sequences: $0 \to \mathbb{Z} \xrightarrow{N} \mathbb{Z}[G] \to \hat{W}_{sep} \to 0$.

Note that δ, δ', δ'' and the right vertical arrow in the third square are isomorphisms when we take p-primary component. By Lemma 3.7 and the above commutative diagram, one obtains

$$[S^2(\hat{T}_{sep})^\Gamma/\mathrm{Dec}(\hat{T}_{sep})]\{p\} \simeq \mathrm{Coker}(H^2(G,\mathbb{Z}) \times H^2(G,\mathbb{Z}) \xrightarrow{\cup} H^4(G,\mathbb{Z}))\{p\}. \quad \square$$

Lemma 3.9. *There is an injective homomorphism $\hat{W}_{sep} \xrightarrow{\varphi} \wedge^2\mathbb{Z}[G]$. And the sequence $0 \longrightarrow \hat{W}_{sep} \xrightarrow{\varphi} \wedge^2\mathbb{Z}[G] \longrightarrow \wedge^2\hat{W}_{sep} \longrightarrow 0$ is exact, where $\wedge^2\mathbb{Z}[G] \longrightarrow \wedge^2\hat{W}_{sep}$ is induced by the surjective homomorphism $\mathbb{Z}[G] \longrightarrow \hat{W}_{sep}$.*

Proof. We first define the homomorphism φ. We write $\bar{g} \in \hat{W}_{sep}$ for the image of $g \in \mathbb{Z}[G]$ and $\bar{g} \wedge \bar{h} \in \wedge^2\hat{W}_{sep}$ for the image of $g \wedge h \in \wedge^2\mathbb{Z}[G]$. Let $\bar{b} \in \hat{W}_{sep}$ and b be an arbitrary lifting of $\bar{b}$. We define $\varphi(\bar{b}) = b \wedge N_G(1)$, and it is easy to see that the morphism φ does not depend on the choice of the lifting.

Then we prove the exactness of the sequence. The surjectivity is obvious. For the exactness at $\wedge^2\mathbb{Z}[G]$, assume that $G = \{g_1, \cdots, g_n\}$, therefore $\wedge^2\hat{W}_{sep}$ has a $\mathbb{Z}$-basis $\{\bar{g}_i \wedge \bar{g}_j\}_{1\le i<j\le n-1}$, since $\bar{g}_n = -\sum_{i=1}^{n-1} \bar{g}_i$ in $\hat{W}_{sep}$. Suppose $a = \sum_{1\le i<j\le n} a_{ij} g_i \wedge g_j \in \wedge^2\mathbb{Z}[G]$ but $\bar{a} = 0 \in \wedge^2\hat{W}_{sep}$. We obtain $\bar{a} = \sum_{1\le i<j\le n-1}(a_{ij} - a_{in} + a_{jn})\bar{g}_i \wedge \bar{g}_j$, since $\bar{g}_n = -\sum_{i=1}^{n-1} \bar{g}_i$ in $\hat{W}_{sep}$, therefore $a_{ij} = a_{in} - a_{jn}$, and $a = \sum_{1\le i<j\le n}(a_{in} - a_{jn}) g_i \wedge g_j$ in $\wedge^2\mathbb{Z}[G]$. Let $b = \sum_{i=1}^{n} a_{in} g_i \in \mathbb{Z}[G]$, then it is clear that $\varphi(\bar{b}) = (\sum_{i=1}^{n} a_{in} g_i) \wedge N_G(1) = a$ and this proves the exactness. $\square$

Combining Theorem 2.1 with Proposition 3.8, we immediately have the following result.

Corollary 3.10. *Let K/F be a Galois extension with $\mathrm{Gal}(K/F) = G$ and $W = R^{(1)}_{K/F}(\mathbb{G}_{m,K})$. Suppose $\#[G,G]$ is a power of 2. Then for an odd prime p,*

$$\bar{H}^3_{nr}(F(W)/F, \mathbb{Q}/\mathbb{Z}(2))\{p\} \simeq H^3(G, K^*)\{p\} \oplus \mathrm{Coker}(H^2(G,\mathbb{Z}) \times H^2(G,\mathbb{Z}) \xrightarrow{\cup} H^4(G,\mathbb{Z}))\{p\}.$$

If G is abelian, then $\#[G,G] = 1$, this satisfies the condition of the above Corollary, and we can give a more precise expression for this cokernel by induction and Hochschild-Serre spectral sequence.

If $G = G' \times G''$ and acts trivially on $\mathbb{Z}$, then the Hochschild-Serre spectral sequence

$$E_2^{p,q} = H^p(G', H^q(G'', \mathbb{Z})) \implies H^{p+q}(G, \mathbb{Z})$$

degenerates, and there is the decomposition by the main theorem of [Jan90]

$$H^n(G, \mathbb{Z}) \simeq \oplus_{p+q=n} H^p(G', H^q(G'', \mathbb{Z})).$$

Moreover, the following diagram is commutative:

$$\begin{array}{ccccc} H^p(G', \mathbb{Z}) & \times & H^q(G'', \mathbb{Z}) & \xrightarrow{\cup} & H^{p+q}(G, \mathbb{Z}) \\ \downarrow \simeq & & \downarrow \simeq & & \downarrow \text{projection} \\ E_2^{p,0} & \times & E_2^{0,q} & \xrightarrow{\cup} & E_2^{p,q} \end{array} \tag{3.5}$$

Since G is finite and abelian, we can represent G as the direct sum of cyclic groups: $G = \oplus_{i=1}^m C_i$ such that $\#C_{i-1} | \#C_i$ $(1 < i \leq m)$. We set $G_i = \oplus_{j=1}^i C_j$, $G = G_m$, $G_{i+1} = G_i \oplus C_{i+1}$.

Lemma 3.11. *Notations as above, let $d_i = (m-i)(m-i-1)/2$, then*

$$\mathrm{Coker}(H^2(G, \mathbb{Z}) \times H^2(G, \mathbb{Z}) \xrightarrow{\cup} H^4(G, \mathbb{Z})) \simeq \oplus_{i=1}^{m-2} G_i^{m-i-1} \simeq \oplus_{i=1}^{m-2} C_i^{d_i}.$$

Proof. Firstly, $H^2(G, \mathbb{Z}) \simeq E_2^{2,0} \oplus E_2^{0,2}$ by the Hochschild-Serre spectral sequence with $G = G_{m-1} \times C_m$ and $E_2^{r,s} = 0$ when s is even. Similarly, $H^4(G, \mathbb{Z}) \simeq E_2^{4,0} \oplus E_2^{2,2} \oplus E_2^{0,4}$. Therefore, from (3.5), we can divide into three steps:

a) $\mathrm{Coker}(E_2^{0,2} \times E_2^{0,2} \xrightarrow{\cup} E_2^{0,4})$: This is trivial by [NSW13, Proposition 1.6.12] since C_m is cyclic.

b) $\mathrm{Coker}(E_2^{2,0} \times E_2^{0,2} \xrightarrow{\cup} E_2^{2,2})$: Assume that $C_m = \mathbb{Z}/n_m\mathbb{Z}$. Since $E_2^{0,2} \simeq C_m = \mathbb{Z}/n_m\mathbb{Z}$, the cokernel of this cup product is equal to the cokernel of the map $H^2(G_{m-1}, \mathbb{Z}) \longrightarrow H^2(G_{m-1}, \mathbb{Z}/n_m\mathbb{Z})$ induced by $\mathbb{Z} \to \mathbb{Z}/n_m\mathbb{Z}$. The exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times n_m} \mathbb{Z} \longrightarrow \mathbb{Z}/n_m \longrightarrow 0$$

derives the exact sequence

$$H^2(G_{m-1}, \mathbb{Z}) \longrightarrow H^2(G_{m-1}, \mathbb{Z}/n_m\mathbb{Z}) \longrightarrow H^3(G_{m-1}, \mathbb{Z}) \xrightarrow{\times n_m} H^3(G_{m-1}, \mathbb{Z}).$$

The last homomorphism is 0 because of the decomposition of G. Thus, by an easy induction, one has $\mathrm{Coker}(E_2^{2,0} \times E_2^{0,2} \xrightarrow{\cup} E_2^{2,2}) \simeq H^3(G_{m-1}, \mathbb{Z}) \simeq \oplus_{i=1}^{m-2} G_i$.

c) $\mathrm{Coker}(E_2^{2,0} \times E_2^{2,0} \xrightarrow{\cup} E_2^{4,0})$: This can be obtained by an induction from the steps a) and b).

Finally, we obtain the result by induction:

$$\begin{aligned}\mathrm{Coker}(H^2(G,\mathbb{Z})\times H^2(G,\mathbb{Z})\xrightarrow{\cup} H^4(G,\mathbb{Z}))&\simeq \oplus_{i=1}^{m-1}H^3(G_i,\mathbb{Z})\\ &\simeq \oplus_{i=1}^{m-2}G_i^{m-i-1}\simeq \oplus_{i=1}^{m-2}C_i^{d_i}.\quad\square\end{aligned}$$

We immediately obtain the following theorem by the previous lemmas.

Theorem 3.12. *Let* $W=R^{(1)}_{K/F}(\mathbb{G}_{m,K})$ *and* K/F *an abelian extension with* $\mathrm{Gal}(K/F)=G\simeq\oplus_{i=1}^{m}C_i$ *such that each* C_i *is cyclic and* $\#C_{i-1}|\#C_i$ $(1<i\leq m)$. *Denote* $d_i=(m-i)(m-i-1)/2$. *Then for any odd prime* p,

$$\bar{H}^3_{nr}(F(W)/F,\mathbb{Q}/\mathbb{Z}(2))\{p\}\simeq H^3(G,K^*)\{p\}\oplus\oplus_{i=1}^{m-2}C_i\{p\}^{d_i}.$$

Moreover, if K/F *has odd degree, then*

$$\bar{H}^3_{nr}(F(W)/F,\mathbb{Q}/\mathbb{Z}(2))\simeq H^3(G,K^*)\oplus\oplus_{i=1}^{m-2}C_i^{d_i}.$$

Proof. By (3.4), if $\mathrm{Gal}(K/F)$ has odd order, 2-primary part of $\bar{H}^3_{nr}(F(W)/F,\mathbb{Q}/\mathbb{Z}(2))$ is trivial, the proof then follows. $\square$

In particular, if $m=2$, we deduce:

Corollary 3.13. *Assume* $W=R^{(1)}_{K/F}(\mathbb{G}_{m,K})$ *and* K/F *an abelian extension with* $\mathrm{Gal}(K/F)=G\simeq C_1\oplus C_2$ *such that* C_1 *and* C_2 *are cyclic groups, then*

$$\bar{H}^3_{nr}(F(W)/F,\mathbb{Q}/\mathbb{Z}(2))\{p\}=H^3(G,K^*)\{p\}$$

for any odd prime p. *In particular, if* G *has odd order, then*

$$\bar{H}^3_{nr}(F(W)/F,\mathbb{Q}/\mathbb{Z}(2))=H^3(G,K^*).$$

Remark. If K/F is a Galois extension of local fields, then $H^3(G,K^*)=0$ by [NSW13, Corollary 7.2.2]. If K/F is a Galois extension of global fields, we consider the exact sequence

$$1\longrightarrow K^*\longrightarrow I_K\longrightarrow C_K\longrightarrow 1$$

where I_K is the idèle group and C_K is the idèle class group. This induces the following commutative diagram with the first row exact [NSW13, Chapter VIII]

$$\begin{array}{ccccccc} H^2(G,I_K) & \xrightarrow{\psi} & H^2(G,C_K) & \longrightarrow & H^3(G,K^*) & \longrightarrow & H^3(G,I_K) \\ \downarrow\simeq & & & & & & \| \\ \oplus_{\mathfrak{p}} H^2(G_{\mathfrak{p}},K_{\mathfrak{p}}^*) & & inv_{K|F}\downarrow\simeq & & & & \oplus_{\mathfrak{p}} H^3(G_{\mathfrak{p}},K_{\mathfrak{p}}^*) \\ inv_{\mathfrak{p}}\downarrow\simeq & & & & & & \| \\ \oplus_{\mathfrak{p}} \frac{1}{[K_{\mathfrak{p}}:F_{\mathfrak{p}}]}\mathbb{Z}/\mathbb{Z} & \longrightarrow & \frac{1}{[K:F]}\mathbb{Z}/\mathbb{Z} & & & & 0 \end{array}$$

where the sum run all prime $\mathfrak{p}$ over F and $G_{\mathfrak{p}}$ is the decomposition subgroup of G with some prime $\mathfrak{B}$ above $\mathfrak{p}$. Therefore $H^3(G,K^*) \simeq \operatorname{coker}\psi$. In general ψ is not surjective, such as $F=\mathbb{Q}$ and $K=\mathbb{Q}(\sqrt{-1},\sqrt{17})$.

Examples. Let $F=\mathbb{Q}(\sqrt{-3})$, K/F an abelian field and $W=R^{(1)}_{K/F}(\mathbb{G}_{m,K})$.

(1) Let $K=F(\sqrt[3]{2},\sqrt[3]{7})$, then $\mathrm{Gal}(K/F)=G\simeq\mathbb{Z}/3\oplus\mathbb{Z}/3$. By the above Remark, one obtains $H^3(G,K^*)=0$ since $K_\nu/F_{\mathfrak{p}}$ has degree 9, where $\nu\mid\mathfrak{p}$ are places over 7. So

$$\bar{H}^3_{nr}(F(W)/F,\mathbb{Q}/\mathbb{Z}(2))=0.$$

(2) Let $K=F(\sqrt[3]{2},\sqrt[3]{3})$, then $\mathrm{Gal}(K/F)=G\simeq\mathbb{Z}/3\oplus\mathbb{Z}/3$. Its local Galois groups are cyclic, which implies $H^3(G,K^*)=\mathbb{Z}/3$ by the above Remark. So

$$\bar{H}^3_{nr}(F(W)/F,\mathbb{Q}/\mathbb{Z}(2))=\mathbb{Z}/3.$$

(3) Let $K=F(\sqrt[3]{2},\sqrt[3]{3},\sqrt[3]{7})$, then $\mathrm{Gal}(K/F)=G\simeq(\mathbb{Z}/3)^3$. Its local Galois groups are $0,\mathbb{Z}/3$ or $(\mathbb{Z}/3)^2$, which implies $H^3(G,K^*)=\mathbb{Z}/3$ by the above Remark. So

$$\bar{H}^3_{nr}(F(W)/F,\mathbb{Q}/\mathbb{Z}(2))=(\mathbb{Z}/3)^2.$$

Data availability

No data was used for the research described in the article.

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